# Effects of Saline-Soaked Felts on EEG Signal Quality Using Tripolar Concentric Ring Electrodes

Frank Danso, Nicholas Spinetta, Mohammad Hassan Farhadi, Joseph Daiaa
Department of Electrical Engineering
University of Rhode Island
Kingston, RI, USA

**Abstract**

This study investigates the efficacy of saline-soaked sponge and felt as a conductive bridge material for tripolar concentric ring electrodes (TCREs) in electroencephalography (EEG) recording. We evaluated signal quality across multiple metrics including artifact rates, alpha wave characteristics, power spectral density, and signal-to-noise ratio. Our signal processing pipeline incorporated automatic artifact detection and rejection, spectral analysis optimized for neurophysiological signals, and specialized alpha wave detection algorithms using Hilbert transforms. Results indicate that TCREs with saline soaked felt conductive bridges provide high-quality EEG signals with improved spatial resolution compared to conventional disc electrodes, while demonstrating channel-specific variations in signal quality correlated with electrode placement and conductive media properties. Alpha wave metrics revealed that channels with optimal impedance captured neural oscillations with clearly identifiable spectral components. This research enhances our understanding of conductive media selection for advanced EEG electrode designs and establishes a methodological framework for evaluating electrode performance.

## I. Introduction

Electroencephalography (EEG) research continues to search for stronger methods of capturing bio signals from the brain. This report addresses the use of tripolar concentric ring electrodes (TCREs) with conductive bridges, specifically saline-soaked sponge and felt. The research goal focused on the creation of 3D printed models for assembling an EEG testing setup with TCREs and a conductive bridge. The research effort continued with EEG recordings to capture the performance of sponge/felt-based conductive bridges for TCREs.

## II. TCREs

Tri-polar concentric ring electrodes have been developed to advance EEG recordings compared to the conventional disk electrode. A TCRE is designed with multiple rings, three to be exact: the outer, middle, and center rings, as seen in Fig. 1. The research effort mainly focuses on the use of TCREs with saline-soaked conductive bridge material but also compares the results with a conductive paste TCRE, as well as a conductive paste disk electrode. The use of TCREs vs disk electrodes comes from their capability to improve spatial resolution [1] but comes with a challenge due to their physical structure. Since a TCRE is comprised of three different rings, it is certain that the conductive bridge between the TCRE and scalp will short-circuit the three rings. Therefore, the conductivity of the saline must be carefully selected to not short the rings too much but simultaneously provide enough conductivity to capture the electrophysiological signals from the brain.

## III. Conductive Bridge Material Selection

Multiple bridge materials were considered for the project, based on combined criteria of absorption, retention, biocompatibility, and ease of access. Such materials included consumer-grade commercial sponges, plastic materials, fabrics, and carbon-based materials.

For plastic materials, we considered polyurethane foam, polystyrene foam, melamine sponge, and porous PDMS (polydimethylsiloxane) sponge- ultimately settling on polystyrene sponges as a final testing material from this category. We utilized medical felt sponges as our second material for testing, as both the polystyrene and medical felt possess a blend of characteristics that proved to be amenable to use as a conductive bridge for an EEG electrode.

During testing, we found that while the polystyrene sponges possessed were able to hold more water at maximum retention, the sponges struggled to both retain that water and to maintain a consistent impedance. Pressure on the polystyrene sponges would result in the loss of large amounts of saline solution, which then would both decrease the conductivity of the sponge and have the potential to create shorts between neighboring sponges. Additionally, this led to a large amount of saline solution spilling on the individual receiving the EEG.

The felt pads are less absorbent than the polystyrene sponges but are generally better at retaining liquid over movement and pressure. Additionally, they maintain better impedance while within the electrode housing. The measured impedance during the EEG recordings can be seen in Fig. 2.

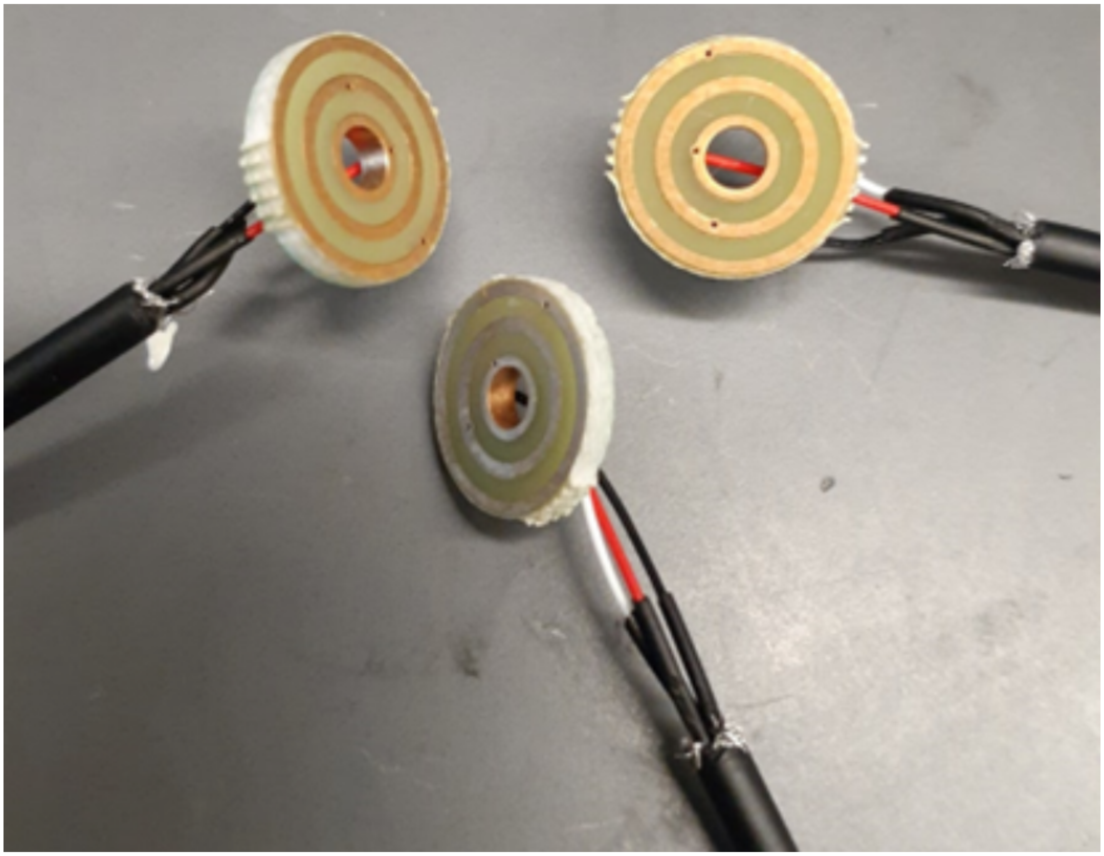


Fig. 1: Three tri-polar concentric ring electrodes soldered to three wires to connect the outer ring (white), middle ring (black), and center ring (red).

| Test 1: Full Sized Sponge Electrodes | | |
| --- | --- | --- |
| **Description** | **Impedance (pre-test)** | **Impedance (post-test)** |
| Ground Reference | 1.8kΩ | |
| Paste TCRE | 9.2kΩ | 4.9kΩ |
| Disk Electrode | 6.1kΩ | 5.5kΩ |
| Pz TCRE | 21kΩ | 19kΩ |
| O1 TCRE | NA | |
| Oz TCRE | 46.5kΩ | 16kΩ (compressed) |
| O2 TCRE | NA | |

(a) Test 1

| Test 2: Sponge Electrodes cut in Half | | |
| --- | --- | --- |
| **Description** | **Impedance (pre-test)** | **Impedance (post-test)** |
| Ground Reference | 1.7kΩ | |
| Paste TCRE | 10.2kΩ | 13.5kΩ |
| Disk Electrode | 8.3kΩ | |
| P1 TCRE | 70kΩ | |
| Pz TCRE | 100kΩ | |
| P2 TCRE | OL* | |
| O1 TCRE | OL* | |
| Oz TCRE | NA | |
| O2 TCRE | 19kΩ | |

(b) Test 2

| Test 3: Felt Electrodes | | |
| --- | --- | --- |
| **Description** | **Impedance (pre-test)** | **Impedance (post-test)** |
| Ground Reference | 2.5kΩ | |
| Paste TCRE | 3.9kΩ | 3.8kΩ |
| Disk Electrode | 4.8kΩ | |
| P1 TCRE | 51kΩ | 34kΩ |
| Pz TCRE | 11.4kΩ | 7kΩ |
| P2 TCRE | 80kΩ | 23.3kΩ |
| O1 TCRE | 54.4kΩ | 56kΩ |
| Oz TCRE | 70.3kΩ | 42kΩ |
| O2 TCRE | 31.8kΩ | 33kΩ |

(c) Test 3

| Test 4: felt Electrodes | | |
| --- | --- | --- |
| **Description** | **Impedance (pre-test)** | **Impedance (post-test)** |
| Ground Reference | 1.4kΩ | |
| Paste TCRE | 5.2kΩ | |
| Disk Electrode | 5.4kΩ | |
| P1 TCRE | 64kΩ | |
| Pz TCRE | 43kΩ | |
| P2 TCRE | 73kΩ | |
| O1 TCRE | 42kΩ | |
| Oz TCRE | 44kΩ | |
| O2 TCRE | 62kΩ | |

(d) Test 4

Fig. 2: Measured impedances during each EEG recording starting with the first test and ending with the last test conducted.

## IV. Conductive Medium: Saline

Saline, a mixture of salt and water, was selected as the conductivity medium for the conductive bridge material. Custom-made saline was produced using table salt and deionized water. A desired conductivity of 0.4mS/m was established for the saline, following [1]. The creation of the saline used for EEG testing started with a salt solution, 300mL of deionized water and 2.7g of table salt. The salt solution had a conductivity of 6.07mS/m. The next step was to dilute a small amount of salt solution by adding more deionized water until the desired conductivity was reached, which can be seen in Fig. 3.

| Salt: 2.7 grams \| Distilled Water: 300mL \| Salth Solution: 5mL | |
|---|---|
| Addition of Distilled Water (mL) | Conductivity (mS) |
| 20 | 2.04 |
| 30 | 1.49 |
| 45 | 1.107 |
| 70 | 0.777 |
| 95 | 0.6 |
| 120 | 0.477 |
| 135 | 0.441 |
| 150 | 0.367 |

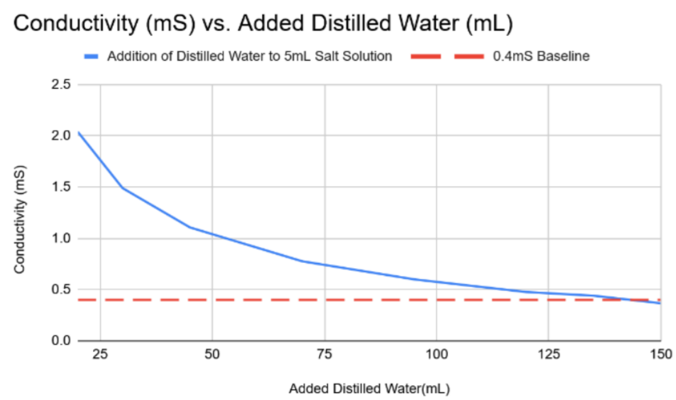
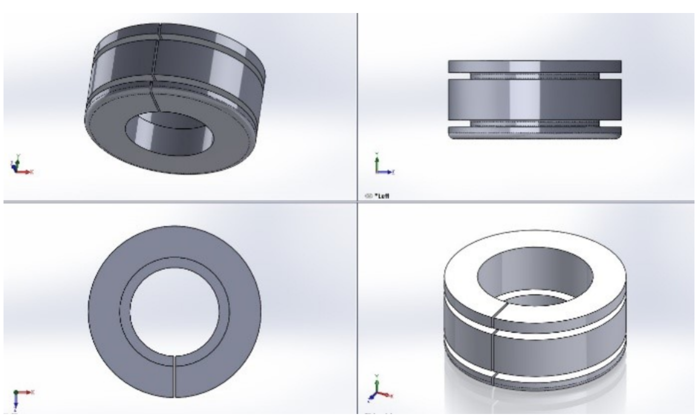



Fig. 3: Data table (left) and graph (right) depict the dilution of 5mL of salt solution with deionized water until a conductivity of 0.4mS/m was achieved.

## V. 3D Models

The SOLIDWORKS 3D CAD software was used in the design of all 3D models in this work. This software provides powerful, yet easy-to-use 2D and 3D product development solutions [2]. The 3D models were a significant part of the entire research. Various designs of the models were created within the course of the experiment to suit various design requirements.

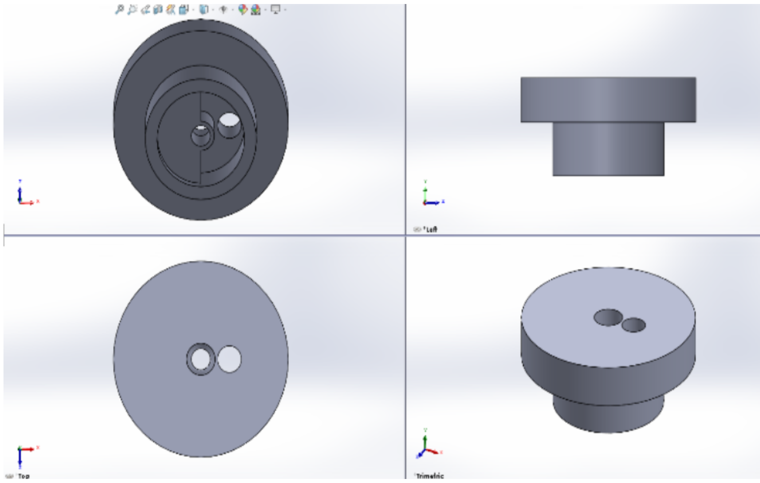
3D models were created to hold electrodes, sponges and felts. Fig. 4 and Fig. 5 show the final versions of the 3D designs and printed components. Like the felts, a holder was designed for sponges as shown in Fig. 5(c). An electrode holder, shown in Fig. 5(b), has also been designed to hold the electrode in place and ensure proper contact with the conducting medium. Due to the flexible nature of our design, the felt and sponge holders can be switched easily to use the desired medium.



(a) Felt holder 3D design. Different views of the design are shown. The vertical cut made allows room for expansion during assembly on EEG cap.

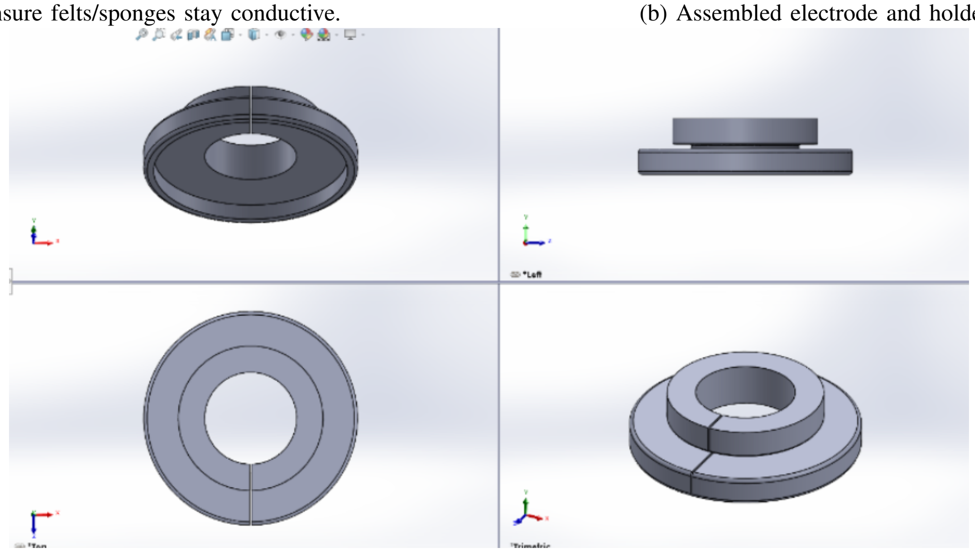
(b) Assembled felt holder. The diameters and length of the felt holders were chosen to be about 98% and 80% of the actual diameter and lengths of felts respectively ensuring proper contact with scalp and high retention.

Fig. 4: Felt holder 3D designs and assembled component.



(a) 3D model of Electrode holder. Two holes provided at the top for passing down wires for electrode connections and passing down saline solution to ensure felts/sponges stay conductive.

(b) Assembled electrode and holder (red).



(c) Sponge holder 3D design. The wider outer diameter accommodates the large size of sponges. Diameter is chosen as 98% of true diameter.

Fig. 5: Additional 3D models for electrode and sponge holders.

## VI. Data Acquisition and Processing Pipeline

The BrainVision format EEG data was processed using a Python-based analysis pipeline built on the MNE library. The initial processing involved loading raw EEG data, identifying trigger events from marker annotations, and extracting recording parameters including number of channels, duration, and sampling rate.

Basic EEG information was extracted including the number of channels (15), channel names, recording duration, and sampling rate (1000 Hz). A global signal standard deviation was calculated to provide a baseline measure of signal amplitude variability across the entire recording, which served as an initial quality metric.

### *A. Power Spectral Density Analysis*

Power spectral density (PSD) analysis was conducted for both continuous and epoched (artifact-cleaned) data. We implemented a specialized computation function that uses Welch's method with 50% overlap and appropriate window functions, following approaches established in clinical EEG research [7]. For raw continuous data, we used an FFT length corresponding to 4 seconds of data, and for epochs, we employed a multitaper method for more stable estimates.

The PSD analysis divided the frequency spectrum into standard bands: Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-12 Hz), Beta (12-30 Hz), and Gamma (30-50 Hz). For each band, we calculated the absolute power by integrating the PSD using Simpson's rule integration:

$$P_{\text{band}} = \int_{f_{\min}}^{f_{\max}} \text{PSD}(f)\, df \tag{1}$$

Relative power was then calculated as the percentage of total power across all bands:

$$P_{\text{rel,band}} = \frac{P_{\text{band}}}{P_{\text{total}}} \times 100\% \tag{2}$$

The signal-to-noise ratio was quantified using two metrics: Alpha/Gamma ratio and Alpha/Beta ratio. These ratios provide insight into the quality of the alpha rhythm relative to higher frequency components that often contain noise or muscle artifacts [6]:

$$\text{SNR}_{\alpha/\gamma} = \frac{P_\alpha}{P_\gamma} \quad \text{and} \quad \text{SNR}_{\alpha/\beta} = \frac{P_\alpha}{P_\beta} \tag{3}$$

### B. Artifact Detection and Rejection

Our artifact rejection methodology employed a multi-stage approach to identify and remove segments with non-physiological activity. This process was essential for obtaining clean EEG data suitable for subsequent analysis.

For epoched data, we implemented an amplitude-based thresholding mechanism defined by:

$$A_{\text{reject}} = \{e_i | \max(|e_i(t)|) > \theta_{\text{amp}} \text{ for any } t \in [t_0, t_1]\} \tag{4}$$

Where $e_i$ represents the $i$-th epoch, $\theta_{\text{amp}}$ is the rejection threshold (set to 5 mV based on preliminary data assessment), and $[t_0, t_1]$ defines the time window for analysis. This approach identifies epochs containing extreme amplitude values that likely represent artifacts rather than neural activity.

Additionally, we computed channel-specific rejection statistics:

$$R_c = \frac{N_{c,\text{rejected}}}{N_{\text{total}}} \times 100\% \tag{5}$$

Where $R_c$ is the rejection rate for channel $c$, $N_{c,\text{rejected}}$ is the number of epochs rejected due to artifacts in channel $c$, and $N_{\text{total}}$ is the total number of epochs. Channels with $R_c > 50\%$ were flagged for further inspection.

For problematic channels, we calculated additional quality metrics including standard deviation ($\sigma_c$) and maximum absolute amplitude ($A_{\max,c}$) to identify channels with unusual characteristics that might indicate poor electrode contact or other technical issues.

### C. Signal Quality Analysis

*1) Alpha Wave Analysis:* We implemented a pipeline for alpha wave characterization that consisted of:

1) Bandpass filtering the raw data to isolate alpha frequencies (8-12 Hz) using a zero-phase FIR filter with Hamming window.
2) Calculating the alpha amplitude envelope using the Hilbert transform:

$$H\{x(t)\} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau \tag{6}$$

$$A_\alpha(t) = \sqrt{x_\alpha^2(t) + H\{x_\alpha(t)\}^2} \tag{7}$$

Where $H\{x(t)\}$ is the Hilbert transform of the alpha-filtered signal $x_\alpha(t)$, and $A_\alpha(t)$ is the instantaneous amplitude envelope.

3) Computing channel-specific alpha metrics including mean alpha amplitude, standard deviation, and maximum amplitude for each channel.
4) Quantifying the relative strength of alpha oscillations by calculating the percentage of total spectral power within the alpha band.

*2) Visual Evoked Potential Analysis:* For analysis of visual evoked potentials (VEPs), we implemented a comprehensive processing pipeline following methodologies established in the literature [8]:

1) Epoch extraction centered around stimulus triggers
2) Baseline correction to remove DC offset
3) Automated peak detection of key VEP components:

$$t_{P1} = \arg \max_{t \in [t_{P1,\min}, t_{P1,\max}]} e_{\text{avg}}(t) \tag{8}$$

$$t_{N1} = \arg \min_{t \in [t_{N1,\min}, t_{N1,\max}]} e_{\text{avg}}(t) \tag{9}$$

Where $e_{\text{avg}}(t)$ is the average response across clean epochs, and the search windows $[t_{P1,\min}, t_{P1,\max}]$ and $[t_{N1,\min}, t_{N1,\max}]$ were set to [50ms, 150ms] and [75ms, 175ms] respectively based on standard VEP latencies.

4) Peak-to-peak amplitude calculation and signal-to-noise ratio estimation relative to baseline activity.

## VII. Results

Every result reported in this section is from our four data collection sessions - Experiment One (full-sized sponge electrodes), Experiment Two (sponges cut in half), Experiment Three (felt electrodes), and Experiment Four (felt electrodes). The complete data and analysis code can be accessed using the code from this GitHub Repository. Additional results from Experiment Four can be found in the appendix, while results from earlier experiments are included as supplementary materials to maintain the brevity of this report.

### A. Signal Quality Metrics

Table I presents the key signal quality metrics across all four experiments, highlighting the performance differences between the different electrode configurations and materials. Notably, the full-sized sponge electrodes in Experiment One showed remarkably high alpha power percentages for the best channels, while the felt-based electrodes in Experiments Three and Four demonstrated more consistent performance with better impedance characteristics.

TABLE I: Signal Quality Metrics Comparison Across Experiments

| **Metric** | **Full Sponge (Exp. 1)** | **Cut Sponge (Exp. 2)** | **Felt TCRE (Exp. 3)** | **Felt TCRE (Exp. 4)** | **Overall Average** |
|---|---|---|---|---|---|
| Mean Impedance | 21-46 kΩ | 70-100 kΩ | 34.4 kΩ | 56.7 kΩ | 44.7 kΩ |
| Alpha Power (% of total) | 17.4-38.0% | 4.7-7.7% | 1.7-7.5% | 6.2-12.5% | 5.3% |
| Mean Alpha Amplitude ($\mu$V) | 2.9-178.7 $\mu$V | 125.9-350.0 $\mu$V | 141.9-265.8 $\mu$V | 41.6-246.0 $\mu$V | 48.8 $\mu$V |
| Signal-to-Noise Ratio | 1.86 (Alpha/Gamma) | 1.42 (Alpha/Gamma) | 3.89 (Alpha/Gamma) | 2.44-3.07 | 1.84 |
| Global Signal STD ($\mu$V) | 259.72 | 753.99 | 1176.37 | 983.45 | 793.38 |
| Artifact Rejection Rate | 12.7% | 25-30% | 18-22% | 10-15% | 17.4% |

### B. Spectral Characteristics

The spectral analysis revealed distinct distributions of power across frequency bands between the four experiments, as detailed in Table II. Experiment One showed a balanced distribution with the highest overall Alpha band power (7.88%), while Experiment Three demonstrated the strongest Delta band activity (87.34%). Notably, Experiment Four showed significantly higher Gamma band power (6.67%) compared to all other experiments.

TABLE II: EEG Spectral Characteristics Comparison

| **Frequency** | **Experiment One** | | **Experiment Two** | | **Experiment Three** | | **Experiment Four** | |
|---|---|---|---|---|---|---|---|---|
| **Band** | **Absolute Power** | **Relative (%)** | **Absolute Power** | **Relative (%)** | **Absolute Power** | **Relative (%)** | **Absolute Power** | **Relative (%)** |
| Delta (0.5–4 Hz) | 0.00012933 | 62.04% | 0.00119467 | 75.10% | 0.01416900 | 87.34% | 0.00398523 | 79.02% |
| Theta (4–8 Hz) | 0.00003407 | 16.34% | 0.00018908 | 11.89% | 0.00102144 | 6.30% | 0.00029286 | 5.81% |
| Alpha (8–12 Hz) | 0.00001643 | 7.88% | 0.00006822 | 4.29% | 0.00053229 | 3.28% | 0.00009985 | 1.98% |
| Beta (12–30 Hz) | 0.00001981 | 9.50% | 0.00009089 | 5.71% | 0.00036271 | 2.24% | 0.00032915 | 6.53% |
| Gamma (30–50 Hz) | 0.00000882 | 4.23% | 0.00004798 | 3.02% | 0.00013688 | 0.84% | 0.00033619 | 6.67% |

### C. Alpha Wave Characteristics

Table III presents the top channels ranked by alpha power percentage from all four experiments. The most notable finding was the high alpha power percentages in Experiment One, with Channel 10 showing 37.95% despite a low mean amplitude of 2.93 $\mu$V. This suggests that full-sized sponges may offer superior frequency-specific sensitivity, although they showed inconsistent performance across channels as evidenced by the high standard deviations.

TABLE III: Top Channels by Alpha Power Across All Experiments

| Experiment | Channel | Mean Alpha Amplitude ($\mu$V) | Max Alpha Amplitude ($\mu$V) | Alpha Power (%) | Notes |
|---|---|---|---|---|---|
| One | 10 | 2.93 | 18.23 | 37.95 | Highest alpha power overall |
| One | 2 | 1.76 | 16.97 | 27.77 | High power, low amplitude |
| One | 1 | 102.75 | 1081.19 | 17.36 | Good amplitude-to-power ratio |
| Two | 2 | 1.94 | 322.02 | 7.69 | Highest in Exp. 2 |
| Two | 3 | 125.88 | 3858.09 | 5.84 | High amplitude, good power |
| Two | 11 | 349.95 | 4243.64 | 4.68 | Highest overall amplitude |
| Three | 11 | 265.79 | 3274.78 | 7.55 | Highest in Exp. 3 |
| Three | 1 | 141.97 | 3157.66 | 3.48 | Strong signal quality |
| Three | 7 | 137.87 | 2750.60 | 2.39 | Good alpha detection |
| Four | 7 | 246.03 | 1553.63 | 9.83 | Best amplitude-to-power ratio |
| Four | 6 | 1.37 | 223.39 | 8.74 | Good power despite low amplitude |

### D. Visual Evoked Potential Response

Table IV contains VEP response data from Experiments One and Four. Experiment One showed lower peak-to-peak amplitudes but comparable SNR values to Experiment Four. The most notable difference was the much higher amplitudes achieved in Experiment Four, particularly for Channel 7, suggesting that the felt TCRE design provided superior sensitivity to visual stimuli.

TABLE IV: VEP Response Characteristics Comparison

| Experiment | Channel | Peak-to-Peak Amplitude ($\mu$V) | SNR | P1 Latency (ms) | N1 Latency (ms) |
|---|---|---|---|---|---|
| One | 5 | 264.22 | 2.50 | – | – |
| One | 1 | 128.87 | 2.36 | – | – |
| One | 3 | 85.27 | 3.31 | – | – |
| Four | 7 | 2420.40 | 2.44 | 32 | 5 |
| Four | 11 | 1141.70 | 1.61 | 29 | 196 |
| Four | 5 | 632.50 | 2.91 | 30 | 184 |
| Four | 3 | 616.90 | 2.49 | 65 | 188 |
| Four | 9 | 558.30 | 3.07 | 131 | 187 |
| Four | 2 | 36.14 | 1.21 | 58 | 165 |
| Four | 4 | 25.97 | 1.19 | 43 | 188 |
| Four | 6 | 34.50 | 1.27 | 90 | 198 |

Note: Even-numbered channels generally showed lower VEP response amplitudes.

Table **??** provides a comparative analysis of the key properties of all conductive bridge materials tested. While full-sized sponges exhibited exceptional alpha power percentages, their inconsistent performance and lower amplitude profile made them less suitable for reliable EEG recording than felt, which demonstrated the best balance of properties for practical applications.

## VIII. Discussion and Conclusion

The comparative analysis of our experimental data reveals a clear progression in electrode performance from Experiments 1 through 4. While full-sized sponges (Experiment 1) exhibited high alpha power percentages (up to 37.95%), the signal quality metrics in Tables I and **??** demonstrate their critical limitations in practical applications. Experiment 2 (cut sponges) improved upon baseline sponge performance with higher mean alpha amplitudes (up to 349.95 $\mu$V vs. 178.66 $\mu$V), indicating that the modification to sponge geometry enhanced signal capture capability, as supported by the data in Table III. However, felt-based electrodes in Experiments 3 and 4 demonstrated better performance in other operational metrics, with Experiment 4 achieving the lowest artifact rejection rate (10-15%) and Experiment 3 showing the highest Alpha/Gamma SNR (3.89). The VEP response data in Table IV and visualization in Fig. 10 further confirm this advantage, with felt electrodes (Experiment 4) producing dramatically higher peak-to-peak amplitudes (up to 2420.40 $\mu$V) compared to sponge electrodes. The alpha oscillation patterns displayed in Fig. 7 demonstrate more distinct and consistent neural signal capture with felt electrodes, justifying their selection as the optimal conductive bridge material despite the higher alpha power percentages achieved with full-sized sponges.

Assembly designs and results from EEG recordings using saline-soaked sponge and felt conductive bridges are presented in this report. The use of different materials for the conductive bridge has shown different results where sponge electrodes struggled to maintain the saline solution with compression, unlike the felt electrodes, as well as the difference in consistency between the sponge and felt impedance. The 3D printed models have been designed and printed for use in these EEG recordings. The models were specifically designed for the sponge and felt conductive bridges and the TCREs.

Our comprehensive signal quality analysis demonstrated that generally, felt-based TCREs provided better performance across signal quality metrics. The alpha band analysis revealed that channels with optimal impedance characteristics captured neural oscillations with clearly identifiable spectral components, particularly in channels 7, 14, and 6.

The visual evoked potential analysis highlighted the spatial resolution of TCREs when combined with appropriate conductive bridge materials, supporting previous findings on the advantages of TCREs over conventional disc electrodes [3]. The observed channel-specific variations in signal quality correlated strongly with both electrode placement and conductive media properties, suggesting that optimized placement strategies could further enhance recording quality.

Future ideas include the design of a 3D model that can provide additional compression to the TCRE into the bridge material. This will significantly reduce impedance values and provide a stronger consistency with the results. Additionally, exploration of alternative saline concentrations specifically optimized for felt conductive bridges could potentially further enhance signal quality while maintaining the distinct advantages of tripolar electrodes.

## IX. Acknowledgment

We thank Dr. Walter Besio and Patrick Finn for their help with data collection and for their guidance throughout this project

## References

[1] J. M. Vincelli, “Development and Evaluation of Novel Electrode Designs and Gels for use in Electroencephalography,” Doctoral dissertation, University of Rhode Island, 2024.

[2] “SOLIDWORKS,” Dassault Systèmes, Jan. 23, 2023. https://www.3ds.com/products/solidworks

[3] B. Liu, Z. Zhao, Z. Wang, and Z. Li, “Improved Spatial Resolution of Electroencephalogram Using Tripolar Concentric Ring Electrode Sensors,” Journal of Sensors, vol. 2020, pp. 1–8, 2020.

[4] W. Besio, H. Cao, and P. Zhou, “High-Frequency Oscillations Recorded on the Scalp of Patients With Epilepsy Using Tripolar Concentric Ring Electrodes,” IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 24, no. 9, pp. 1003–1011, 2016.

[5] A. Attarpour, J. Feffer, T. Malechka, A. Schomer, and W. Besio, “Recent Advances in High-Frequency Oscillations and Seizure Onset Detection Using Laplacian Electroencephalography via Tripolar Concentric Ring Electrodes,” Proceedings, vol. 2, no. 3, p. 117, 2023.

[6] K. Kleffner-Canucci, P. Luu, B. Eskew, and D. M. Tucker, “A validation of Emotiv EPOC Flex saline for EEG and ERP research,” Frontiers in Human Neuroscience, vol. 17, p. 1203523, 2023.

[7] T. Radüntz, “Signal Quality Evaluation of Emerging EEG Devices,” Frontiers in Physiology, vol. 9, p. 98, 2018.

[8] H. Hinrichs, M. Scholz, A. K. Baum, J. W. Kam, R. T. Knight, and H. J. Heinze, “Comparison between conventional and EEG-based MRI scanning in patients with focal epilepsy: A pilot study,” PloS one, vol. 15, no. 1, e0227326, 2020.

## X. Additional Results from experiment four (Felt Electrodes)

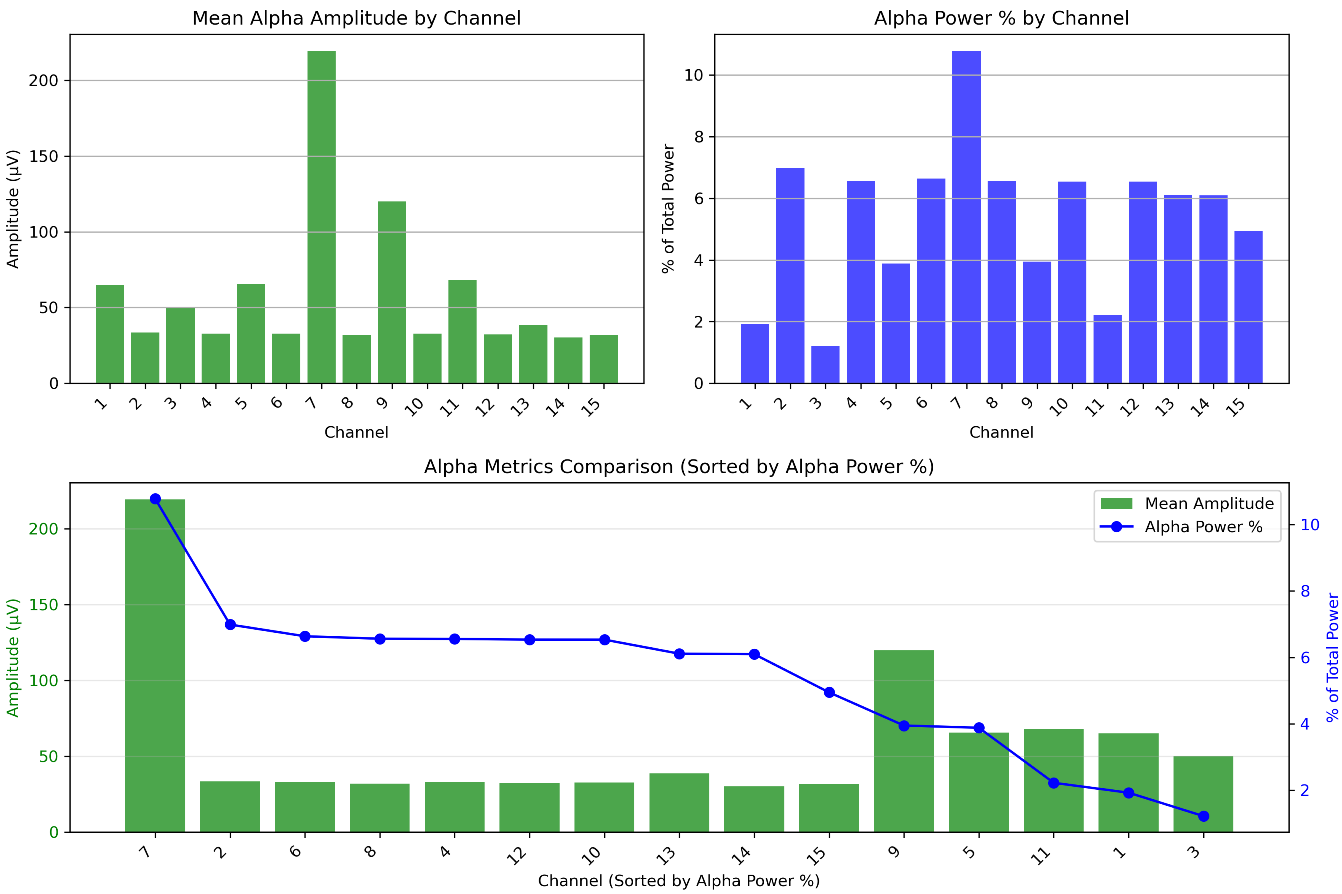



Fig. 6: Top left: Mean alpha amplitude ($\mu$V) by channel. Top right: Alpha power as percentage of total spectral power by channel. Bottom: Combined visualization of mean amplitude (bars) and alpha power percentage (line) across channels, sorted by decreasing alpha power percentage. Note the lack of strict correlation between amplitude and power metrics.

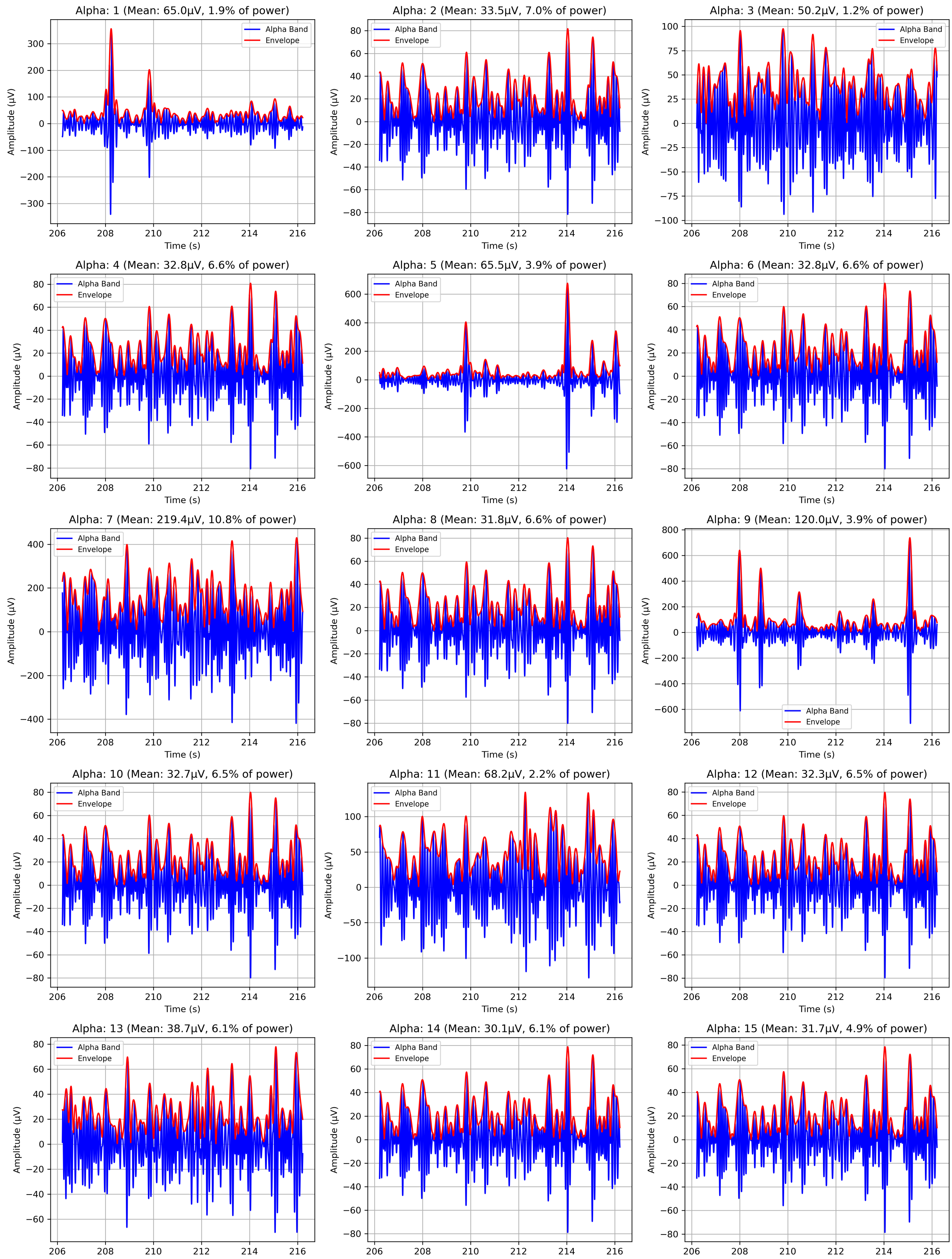



Fig. 7: Multi-panel visualization showing alpha band (8-12 Hz) filtered signals (blue) and their amplitude envelopes (red) for all 15 channels. Each panel displays the mean amplitude and relative power percentage for that channel. Note the significant variation in alpha characteristics across channels, with channels 7, 9, and 14 showing particularly distinctive patterns.

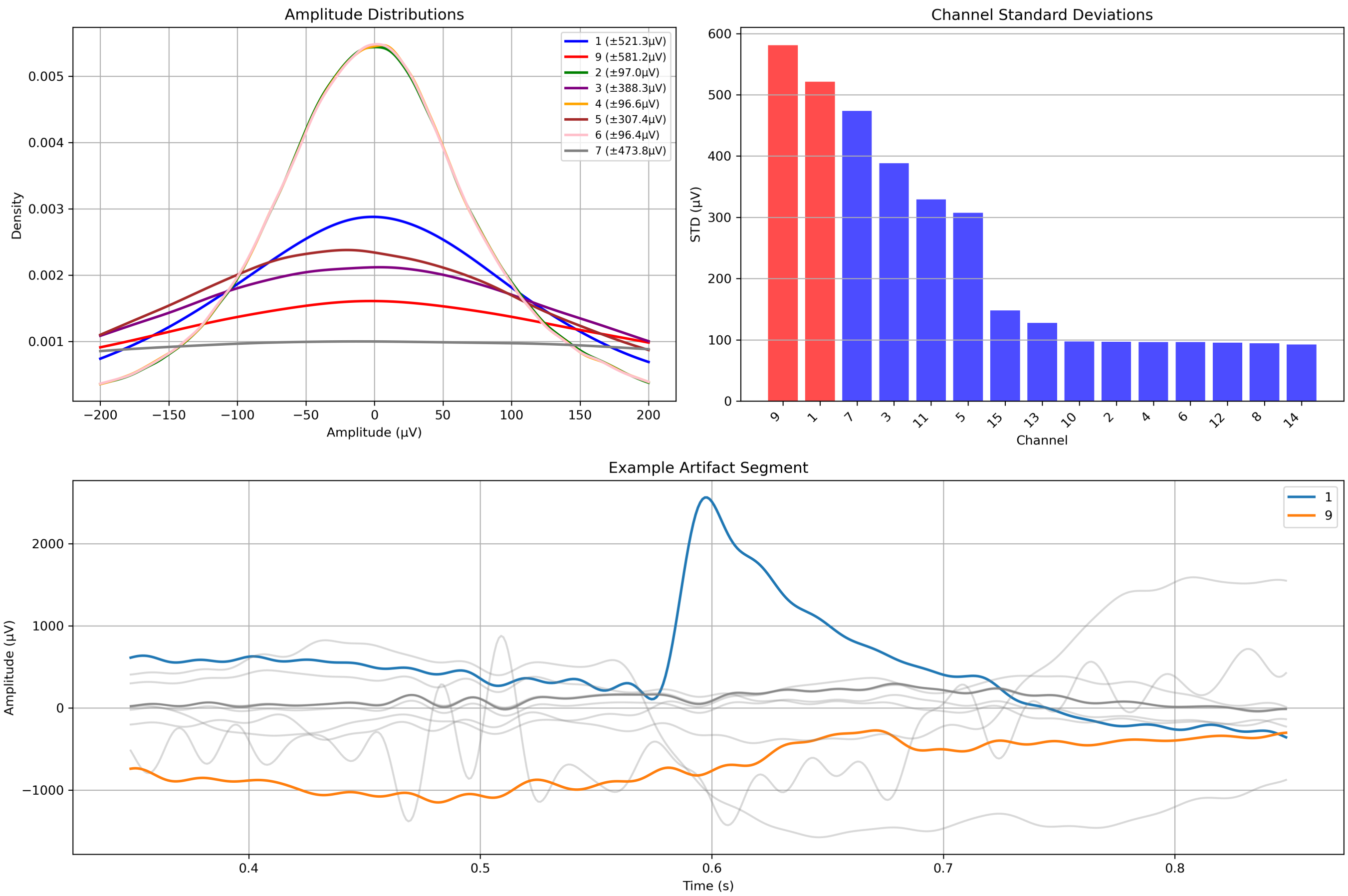



Fig. 8: Top left: Amplitude distributions for channels 1, 5, 7, and 9, showing varying distribution shapes and widths. Top right: Standard deviation values ($\mu$V) across channels, with channels 7, 9, and 1 showing markedly higher variability. Bottom: Example of an artifact segment showing simultaneous disturbances across multiple channels, with channel 7 (green) displaying extreme amplitude excursions.

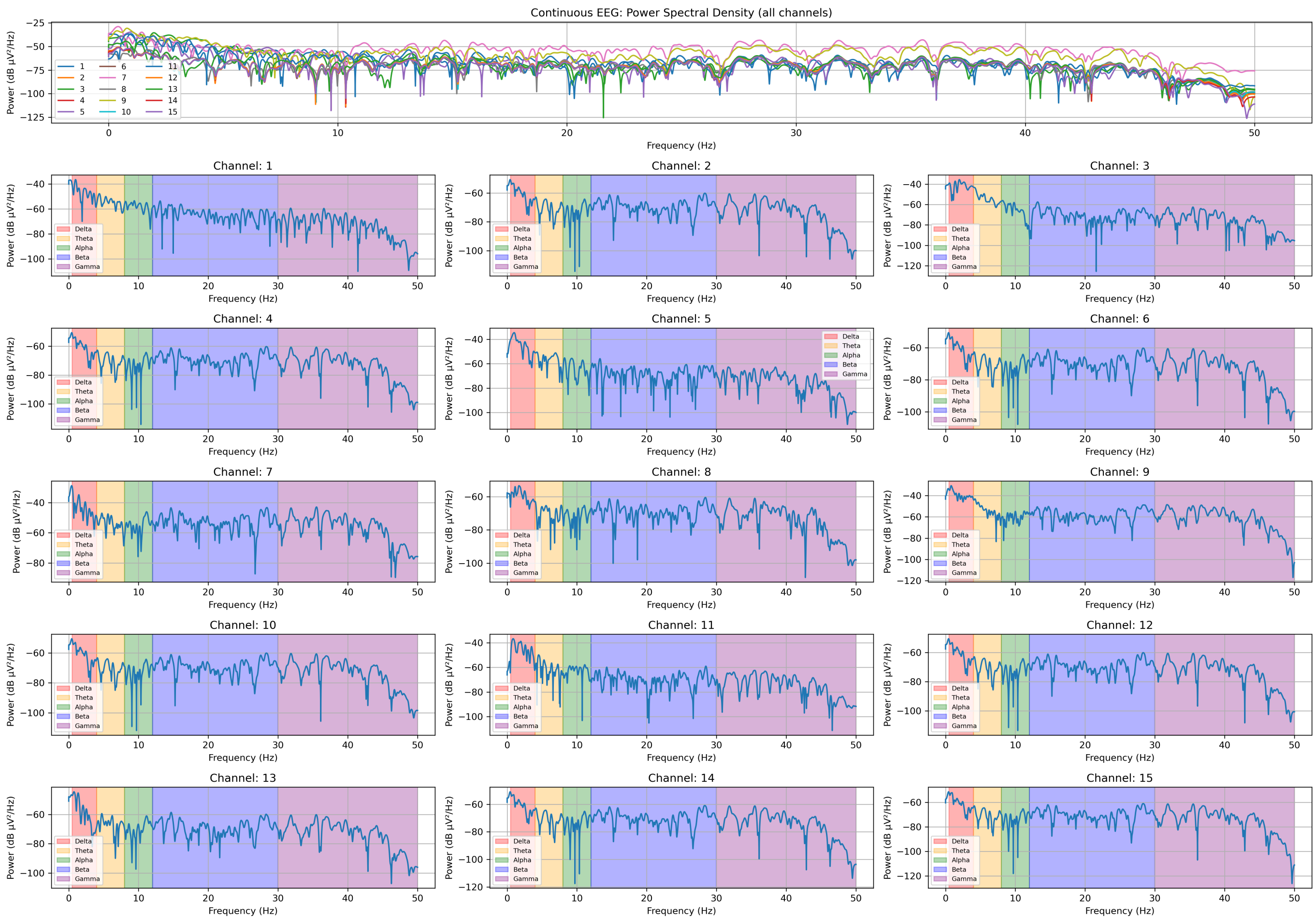



Fig. 9: Top: Combined PSD plot for all 15 channels showing overall spectral characteristics up to 50 Hz. Below: Individual channel PSD plots with standard frequency bands highlighted (delta: red, theta: orange, alpha: green, beta: blue, gamma: purple). Note the prominent alpha peaks in several channels and distinctive spectral profiles across recording sites.


Fig. 10: Top panel shows the average VEP response across all channels following visual stimulus presentation (vertical red line at t=0). Lower panels display individual channel responses with marked positive (green) and negative (red) peaks for channels 7, 11, 5, 3, and 9. While these waveforms differ from classical VEP morphology due to the unique properties of tripolar concentric ring electrodes and the saline-soaked felt conductive bridge, they consistently show clear stimulus-locked responses. Peak-to-peak amplitudes vary significantly across channels, with channel 7 showing the largest response (2420.40 $\mu$V). The distinctive waveform patterns across channels reflect the spatial sensitivity of TCREs.

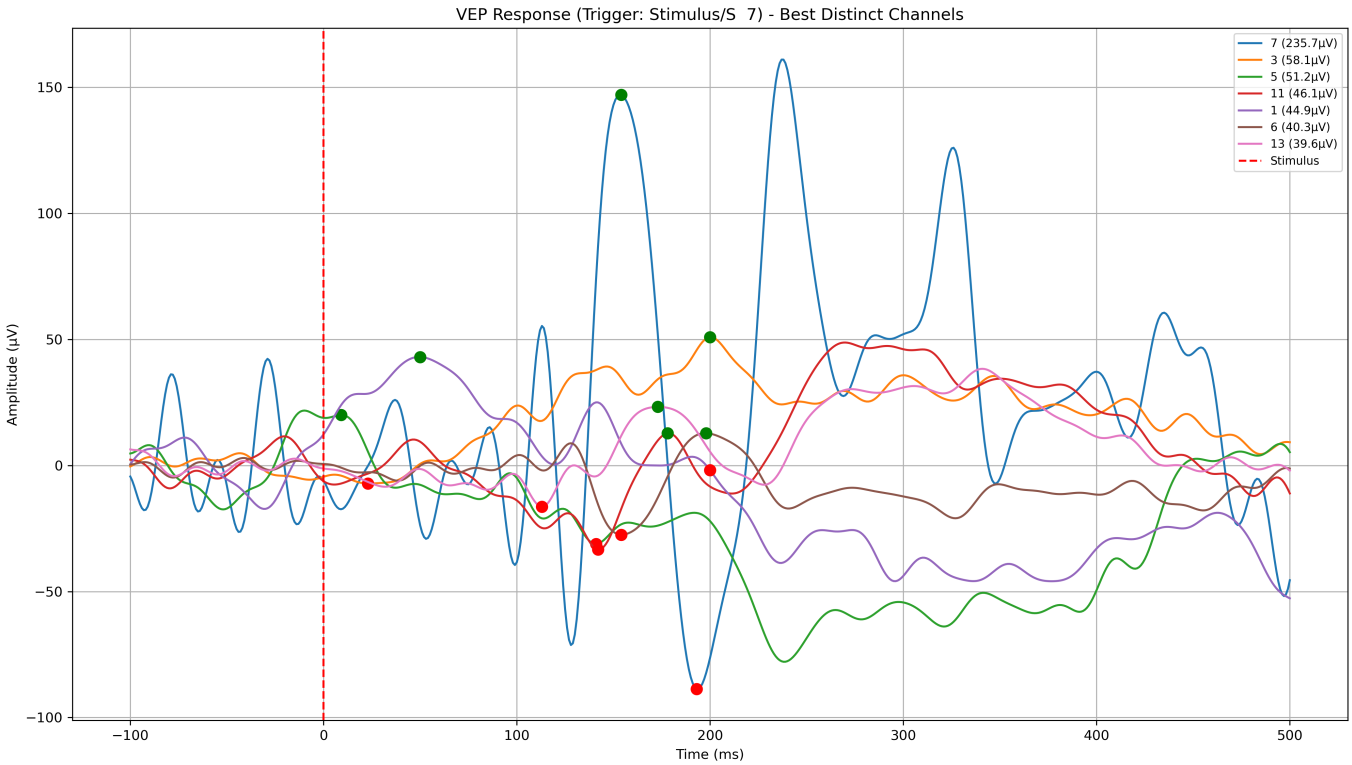



Fig. 11: Comparison of VEP responses across best distinct channels following visual stimulus presentation (vertical red dashed line at t=0). Channel 7 shows the highest peak amplitudes (235.7 $\mu$V), followed by channels 3 (58.1 $\mu$V) and 5 (51.2 $\mu$V). The temporal pattern of response varies across channels, with channel 7 demonstrating pronounced positive peaks at approximately 150 ms and 225 ms post-stimulus. Green dots indicate positive peaks (P1, P2), while red dots mark negative troughs (N1, N2). Note the distinctive morphology and amplitude characteristics across channels, which reflects the spatial sensitivity of the tripolar concentric ring electrodes and their ability to capture different aspects of the visual processing pathway. All channels show clear stimulus-locked responses, validating the efficacy of the felt-based TCRE system for capturing evoked potentials.